Minimal 1448 airboat electrical schematic: power distribution

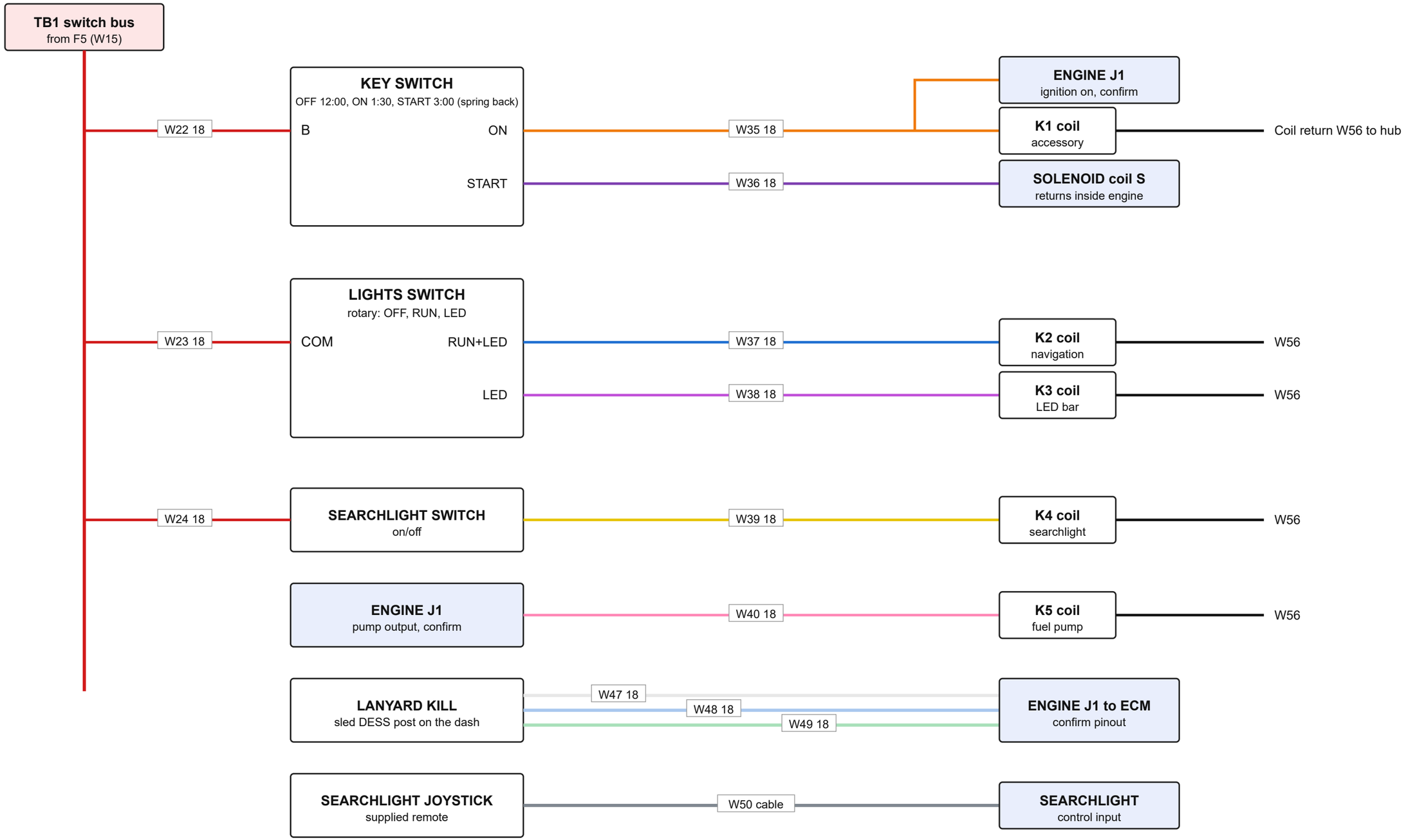
Generated from electrical/wire_list.py. Red is power, black is return, other colors are signals or sensors. Chassis ground only inside the engine.



Relay contacts sit in the dry box. Coils are on sheet 2.

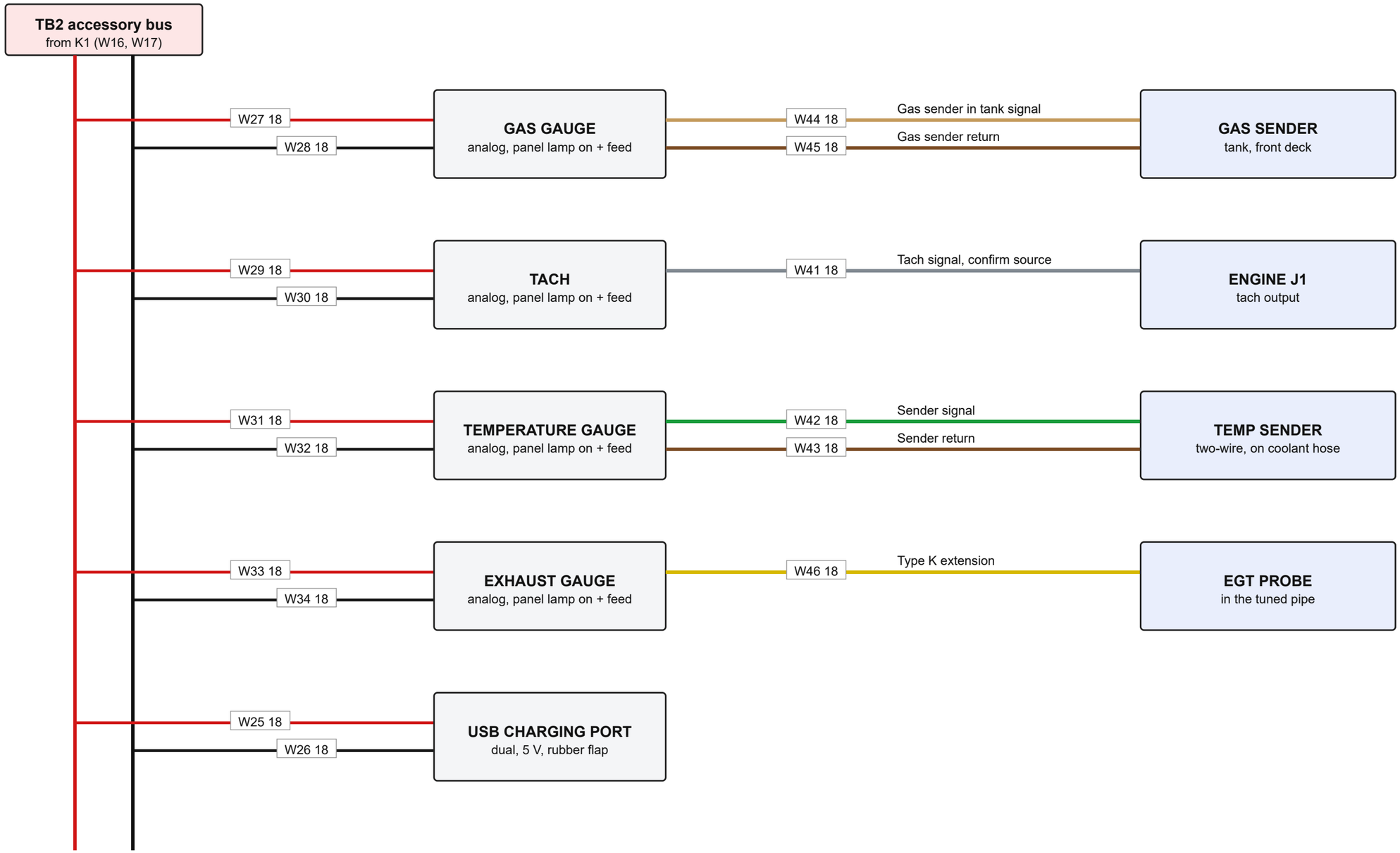
Minimal 1448 airboat electrical schematic: switching and signals

Generated from electrical/wire_list.py. Red is power, black is return, other colors are signals or sensors. Chassis ground only inside the engine.



Minimal 1448 airboat electrical schematic: gauges and sensors

Generated from electrical/wire_list.py. Red is power, black is return, other colors are signals or sensors. Chassis ground only inside the engine.



Minimal 1448 airboat electrical schematic: wire and fuse lists

Generated from electrical/wire_list.py. Red is power, black is return, other colors are signals or sensors. Chassis ground only inside the engine.

Wire	Function	AWG	Color	Amps	Ft	Drop
W01	Battery positive to starter solenoid	1/0	red	150	6.6	
W02	Battery negative to engine ground	1/0	black	150	7.1	
W03	Battery positive to hub	6	red	45	0.7	
W04	Battery negative to hub	6	black	45	0.8	
W05	F1 to 12 V outlet	12	red	20	4.6	2.1%
W06	12 V outlet return	12	black	20	4.6	
W07	K3 to LED bar	12	red	10.3	16.9	4.0%
W08	LED bar return	12	black	10.3	16.7	
W09	K4 to searchlight	18	red	3.7	17.2	5.9%
W10	Searchlight return	18	black	3.7	17.0	
W11	K2 to stern light	18	red	0.5	17.7	0.8%
W12	Stern light return	18	black	0.5	17.5	
W13	K2 to bow light	18	red	0.5	9.9	0.5%
W14	Bow light return	18	black	0.5	9.9	
W15	F5 to dash switch bus	18	red	6	4.5	2.5%
W16	K1 to dash accessory bus	18	red	3.5	4.9	1.6%
W17	Dash accessory return	18	black	3.5	4.6	
W18	F7 to bilge pump	18	red	2	7.2	1.3%
W19	Bilge pump return	18	black	2	7.2	
W20	K5 to fuel pump	12	red	6	9.3	1.3%
W21	Fuel pump return	12	black	6	9.3	
W22	Switch bus to key	18	red	6	0.8	0.4%
W23	Switch bus to lights switch	18	red	1	1.0	0.1%
W24	Switch bus to searchlight switch	18	red	1	1.2	0.1%
W25	Accessory bus to USB	18	red	1.5	1.5	0.2%
W26	USB return	18	black	1.5	1.5	
W27	Accessory bus to gas gauge	18	red	0.3	1.3	0.0%
W28	Gas gauge return	18	black	0.3	1.3	
W29	Accessory bus to tach	18	red	0.3	1.0	0.0%
W30	Tach return	18	black	0.3	1.0	
W31	Accessory bus to temperature gauge	18	red	0.3	0.8	0.0%
W32	Temperature gauge return	18	black	0.3	0.8	
W33	Accessory bus to exhaust gauge	18	red	0.3	0.7	0.0%
W34	Exhaust gauge return	18	black	0.3	0.7	
W35	Key ON to K1 coil and engine	18	orange	0.5	11.2	
W36	Key START to solenoid	18	purple	5	11.0	
W37	Lights switch RUN to K2 coil	18	blue	0.2	5.2	
W38	Lights switch LED to K3 coil	18	violet	0.2	5.2	
W39	Searchlight switch to K4 coil	18	yellow	0.2	5.4	
W40	Engine pump signal to K5 coil	18	pink	0.2	6.9	
W41	Tach signal	18	gray	0.05	11.5	
W42	Temperature sender signal	18	green	0.05	12.3	
W43	Temperature sender return	18	brown	0.05	12.3	
W44	Gas sender signal	18	tan	0.05	13.6	
W45	Gas sender return	18	brown	0.05	13.6	
W46	Exhaust probe, type K extension	18	K yellow	0.0	14.1	
W47	DESS post to ECM, wire 1	18	white	0.05	12.0	
W48	DESS post to ECM, wire 2	18	white/blue	0.05	12.0	
W49	DESS post to ECM, wire 3	18	white/green	0.05	12.0	
W50	Searchlight remote cable	18	gray	0.1	22.1	
W51	Hub fuse to relay K1 accessory	18	red	3.5	0.9	0.3%
W52	Hub fuse to relay K2 navigation	18	red	1	0.9	0.1%
W53	Hub fuse to relay K3 LED bar	12	red	10.3	0.9	0.2%
W54	Hub fuse to relay K4 searchlight	18	red	3.7	0.9	0.3%
W55	Hub fuse to relay K5 fuel pump	12	red	6	0.9	0.1%
W56	Relay coil returns (5, daisy chain)	18	black	1	0.9	

Fuses

MF1 60 A Main feed, at the battery (W03)
F1 20 A 12 V outlet (W05)
F2 15 A LED bar (K3) (W07)
F3 10 A Searchlight (K4) (W09)
F4 10 A Navigation lights (K2) (W11, W13)
F5 10 A Dash switch bus: key, lights, searchlight (W15)
F6 10 A Accessory bus: gauges, USB (K1) (W16)
F7 10 A Bilge pump (W18)
F8 15 A Fuel pump (K5) (W20)

To confirm from the 800R service manual

Engine J1 pinout: ignition on, tach, fuel pump output, DESS post.
Whether the ECM runs the pump on and off, which K5 assumes, or varies it.
Fuel pump current. 6 A is an estimate.
Charging output and where the regulator lands.
Solenoid coil current. 5 A is an estimate.
Whether the kept DESS post works as the lanyard kill.

Connectors

Battery, solenoid, ground stud, fuse block: sealed ring terminals, adhesive heat shrink.
Relays and dash switches: sealed 1/4 in female spades.
Devices outside the dry box: Deutsch DT, 2-pin for a device pair. Engine J1: Deutsch DT 12-pin on the stand.
Gauges: ring terminals on the gauge studs. Exhaust probe: type K mini plug.